


# ACADEMIC SESSION 2026-2027

## LAB PRACTICAL FILE

# DATA STRUCTURE

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**course : MCA**

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# Program 1

**AIM :** Insert a new element in array

practice.py

```python

arr = []

n = int(input("Enter number of elements: "))

for i in range(n):
    x = int(input("Enter element: "))
    arr.append(x)

print("Original array:", arr)

element = int(input("Enter element to insert: "))
position = int(input("Enter position: "))

arr.insert(position, element)

print("Array after insertion:", arr)

```

## OUTPUT

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter number of elements: 5
Enter element: 12
Enter element: 14
Enter element: 16
Enter element: 18
Enter element: 20
Original array: [12, 14, 16, 18, 20]
Enter element to insert: 50
Enter position: 2
```

# Program 2

**AIM :** Delete an element in array

practice.py

```python
arr = []

n = int(input("Enter number of elements: "))

for i in range(n):
    x = int(input("Enter element: "))
    arr.append(x)

print("Original array:", arr)

element = int(input("Enter element to delete: "))

if element in arr:
    arr.remove(element)
    print("Array after deletion:", arr)
else:
    print("Element not found")
```

## OUTPUT

Problems Output Debug Console Terminal Ports

```
Enter number of elements: 5
Enter element: 10
Enter element: 20
Enter element: 30
Enter element: 40
Enter element: 50
Original array: [10, 20, 30, 40, 50]
Enter element to delete: 30
Array after deletion: [10, 20, 40, 50]
PS C:\Users\hp\OneDrive\Desktop\python>
```

# Program 3

**AIM :** Find product of two matrix

practice.py

```python

r = int(input("Enter number of rows: "))
c = int(input("Enter number of columns: "))

A = []
B = []

print("Enter elements of first matrix:")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    A.append(row)

print("Enter elements of second matrix:")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    B.append(row)

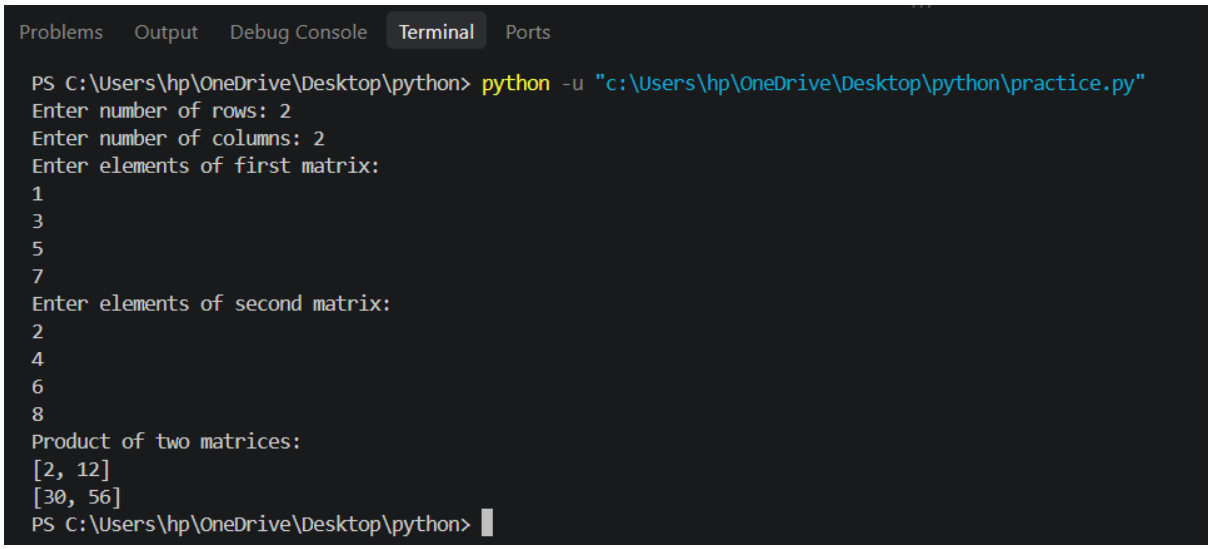
C = []

for i in range(r):
    row = []
    for j in range(c):
        row.append(A[i][j] * B[i][j])
    C.append(row)

print("Product of two matrices:")
for row in C:
    print(row)

```

OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter number of rows: 2
Enter number of columns: 2
Enter elements of first matrix:
1
3
5
7
Enter elements of second matrix:
2
4
6
8
Product of two matrices:
[2, 12]
[30, 56]
PS C:\Users\hp\OneDrive\Desktop\python>

# Program 4

**AIM :** Sum of diagonal and non diagonal elements in matrix

practice.py

```python

n = int(input("Enter size of matrix: "))

A = []

print("Enter matrix elements:")
for i in range(n):
    row = []
    for j in range(n):
        row.append(int(input()))
    A.append(row)

diagonal_sum = 0
non_diagonal_sum = 0

for i in range(n):
    for j in range(n):
        if i == j:
            diagonal_sum += A[i][j]
        else:
            non_diagonal_sum += A[i][j]

print("Diagonal sum:", diagonal_sum)
print("Non-diagonal sum:", non_diagonal_sum)

```

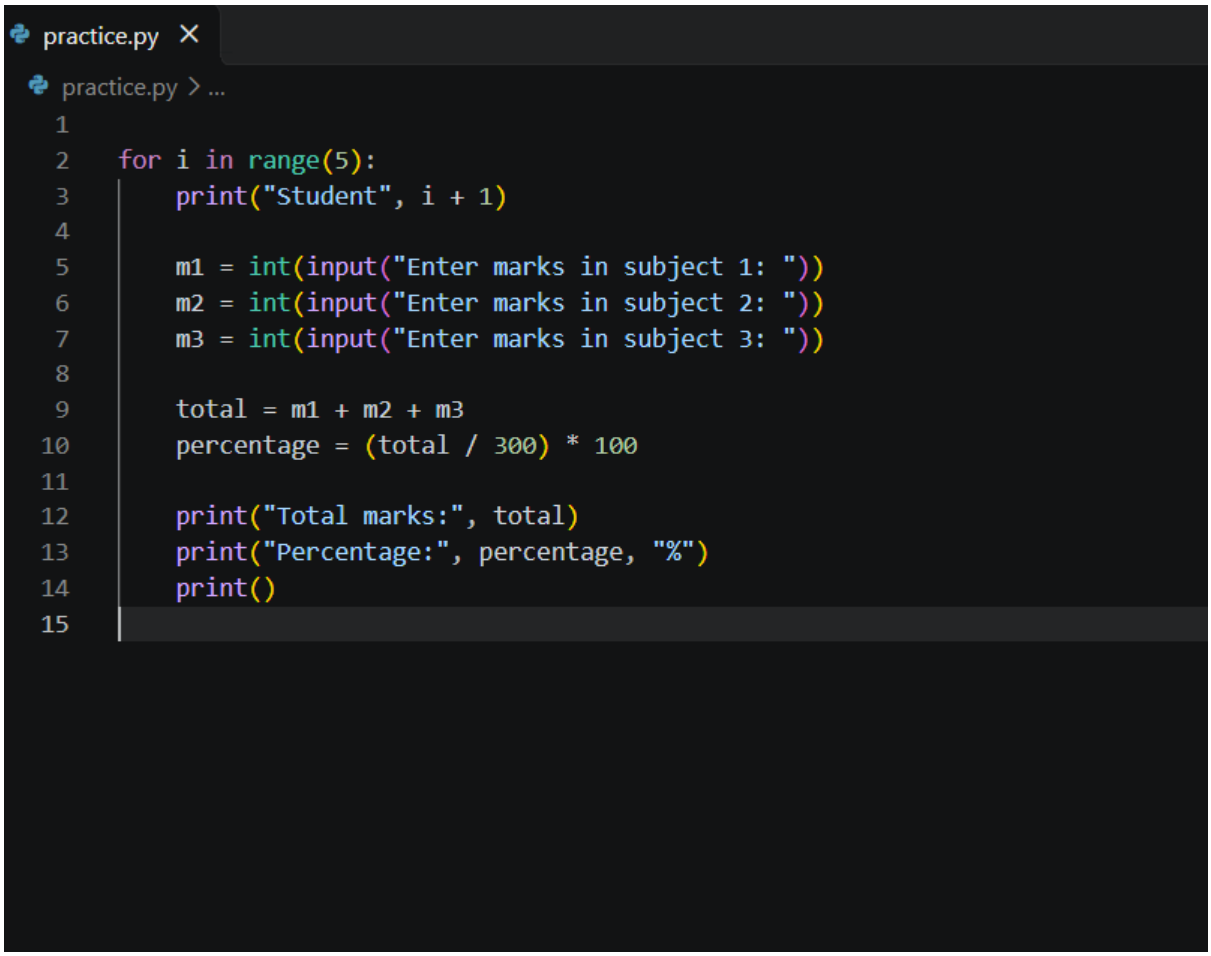
## OUTPUT

```
Problems  Output  Debug Console  Terminal  Ports

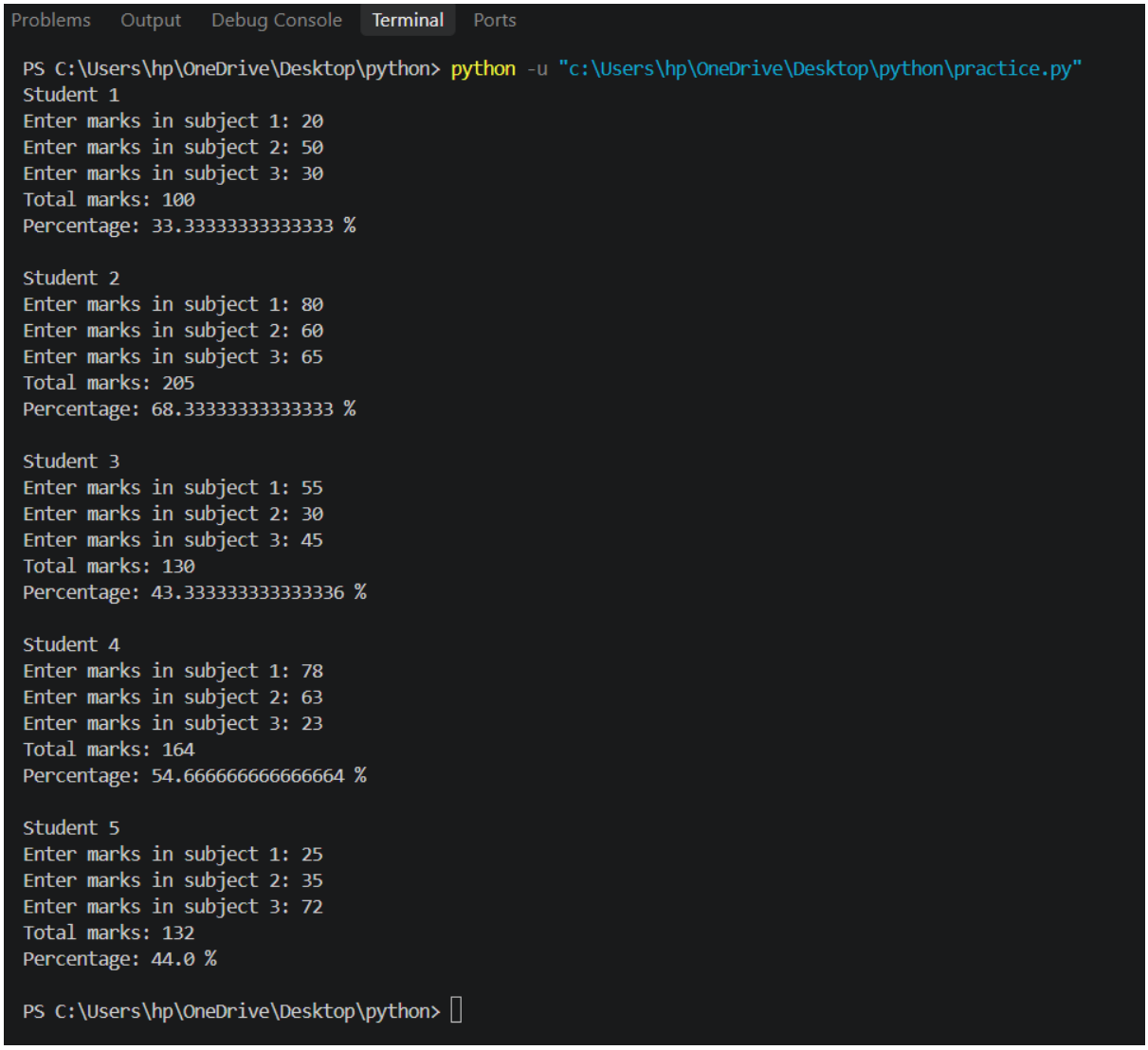
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter size of matrix: 2
Enter matrix elements:
1
2
3
4
Diagonal sum: 5
Non-diagonal sum: 5
PS C:\Users\hp\OneDrive\Desktop\python>
```

# Program 5

**AIM :** Enter marks of 5 students for 3 subjects each and find the percentage of students



# OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Student 1
Enter marks in subject 1: 20
Enter marks in subject 2: 50
Enter marks in subject 3: 30
Total marks: 100
Percentage: 33.33333333333333 %
Student 2
Enter marks in subject 1: 80
Enter marks in subject 2: 60
Enter marks in subject 3: 65
Total marks: 205
Percentage: 68.33333333333333 %
Student 3
Enter marks in subject 1: 55
Enter marks in subject 2: 30
Enter marks in subject 3: 45
Total marks: 130
Percentage: 43.333333333333336 %
Student 4
Enter marks in subject 1: 78
Enter marks in subject 2: 63
Enter marks in subject 3: 23
Total marks: 164
Percentage: 54.666666666666664 %
Student 5
Enter marks in subject 1: 25
Enter marks in subject 2: 35
Enter marks in subject 3: 72
Total marks: 132
Percentage: 44.0 %
PS C:\Users\hp\OneDrive\Desktop\python>

# Program 6

**AIM :** Stack-

1. opertaion
2. reverse of string
3. parenthesis Matching

## 1. operation

practice.py

```python
stack = []

while True:
    print("\n
    1. Push")
    print("2. Pop")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        x = int(input("Enter element: "))
        stack.append(x)
        print("Element inserted")

    elif choice == 2:
        if len(stack) == 0:
            print("Stack is empty")
        else:
            print("Deleted element:", stack.pop())

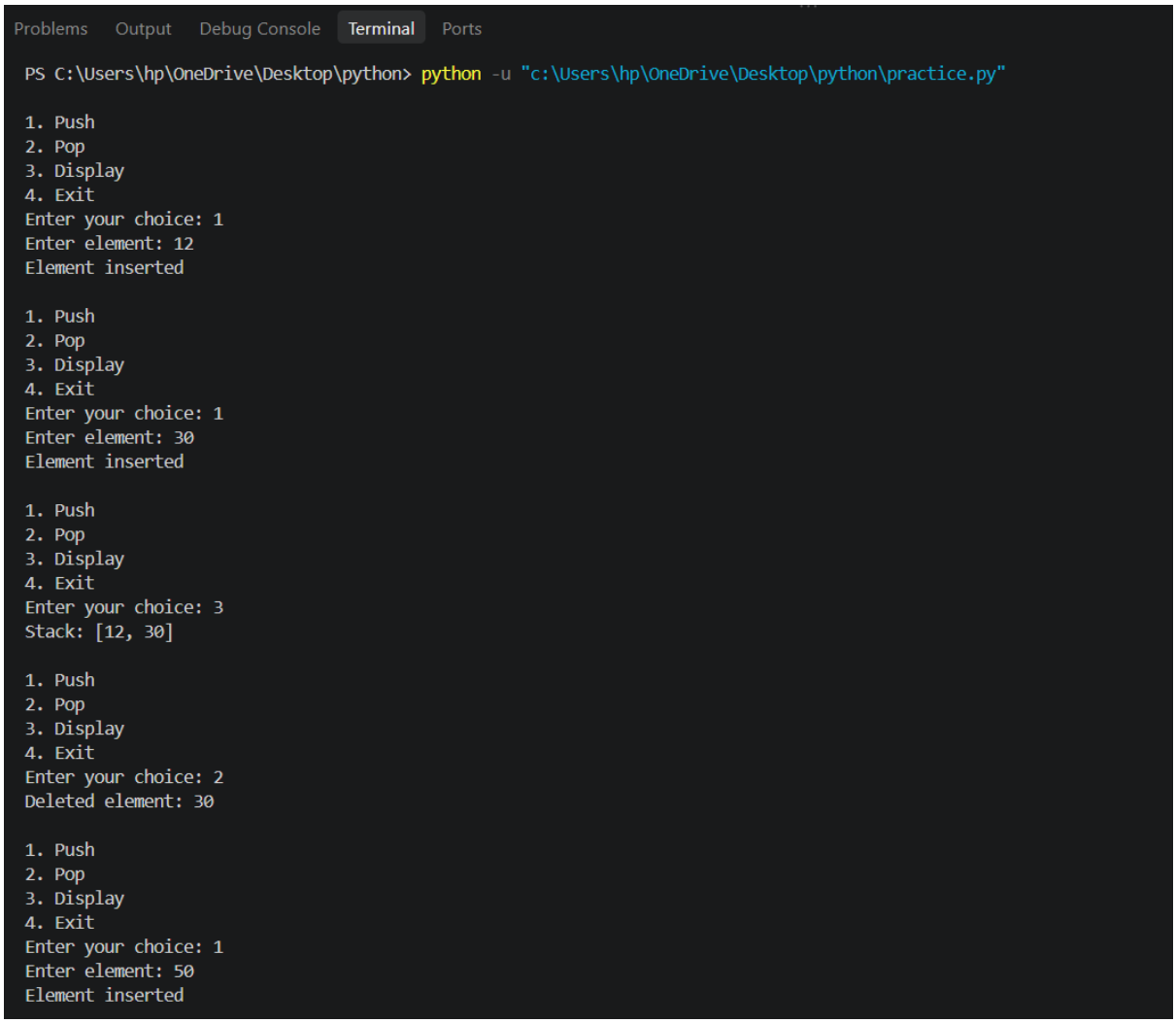
    elif choice == 3:
        print("Stack:", stack)

    elif choice == 4:
        break

    else:
        print("Invalid choice")

```

# OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element: 12
Element inserted
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack: [12, 30]
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2
Deleted element: 30
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter element: 50
Element inserted

# 2. reverse of string

practice.py

```python

stack = []

s = input("Enter a string: ")

for ch in s:
    stack.append(ch)

rev = ""

while len(stack) > 0:
    rev = rev + stack.pop()

print("Reversed string:", rev)

```

## OUTPUT



# 3. parenthesis Matching

practice.py

```python

stack = []

s = input("Enter an expression: ")

for ch in s:
    if ch == '(' or ch == '{' or ch == '[':
        stack.append(ch)

    elif ch == ')' or ch == '}' or ch == ']':
        if len(stack) == 0:
            print("Parentheses are not balanced")
            break

        top = stack.pop()

        if (ch == ')' and top != '(') or \
           (ch == '}' and top != '{') or \
           (ch == ']' and top != '['):
            print("Parentheses are not balanced")
            break
else:
    if len(stack) == 0:
        print("Parentheses are balanced")
    else:
        print("Parentheses are not balanced")

```

## OUTPUT

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter an expression: (a+b)*(a-b)
Parentheses are balanced
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter an expression: (a+b]
Parentheses are not balanced
PS C:\Users\hp\OneDrive\Desktop\python>
```

# Program 5

**AIM :** Queue-

1.general Queue Operations

2. Circular Queue Operations

1.general Queue Operations

practice.py

```python
queue = []

while True:
    print("\n
    1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        x = int(input("Enter element: "))
        queue.append(x)
        print("Element inserted")

    elif choice == 2:
        if len(queue) == 0:
            print("Queue is empty")
        else:
            print("Deleted element:", queue.pop(0))

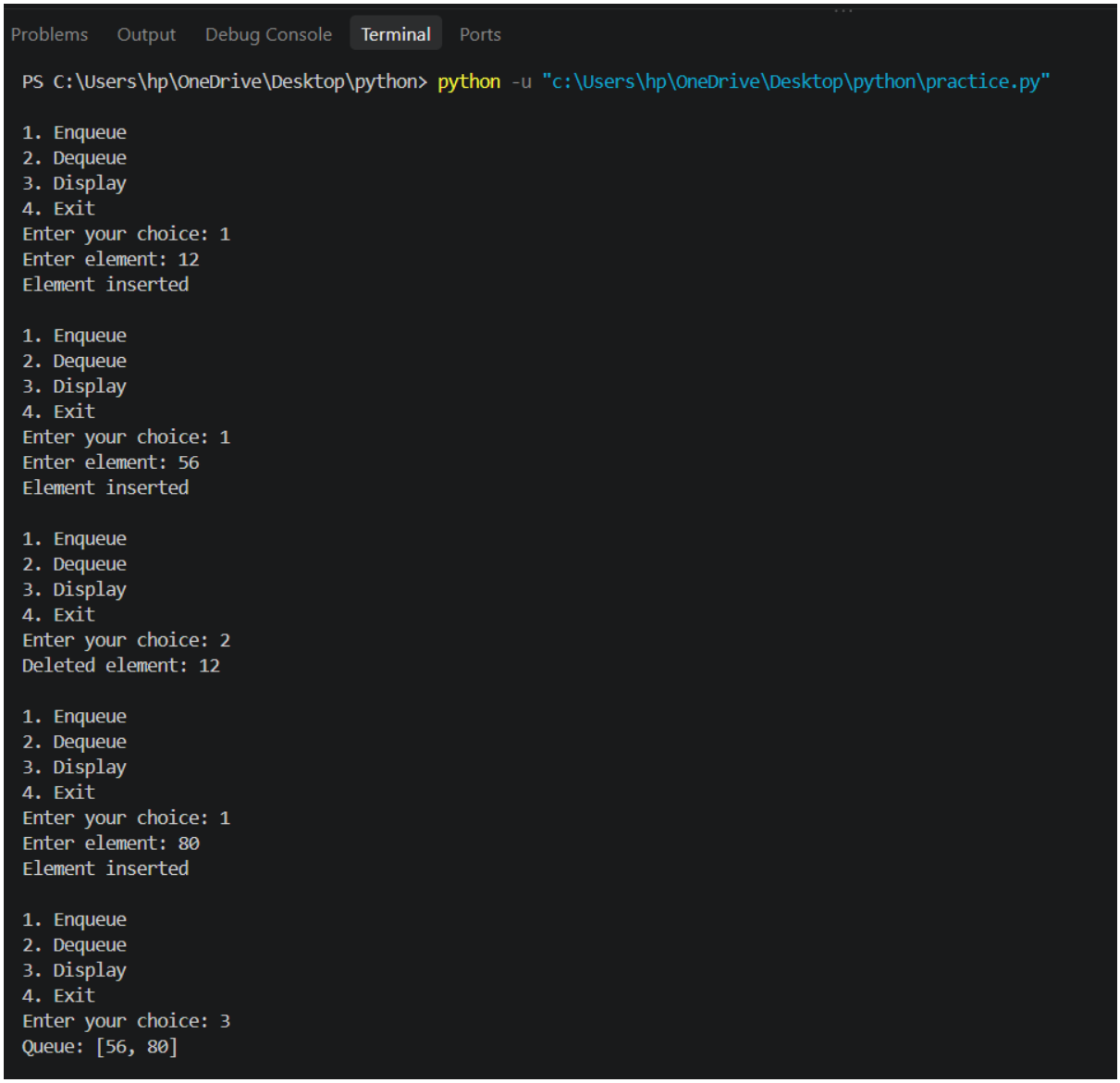
    elif choice == 3:
        print("Queue:", queue)

    elif choice == 4:
        break

    else:
        print("Invalid choice")

```

# OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 12
Element inserted
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 56
Element inserted
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted element: 12
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 80
Element inserted
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue: [56, 80]

# 2. Circular Queue Operations

practice.py

```python
queue = []
size = int(input("Enter size of queue: "))
front = -1
rear = -1
while True:
    print("\n
    1. Enqueue")
    print("2. Dequeue")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        if (rear + 1) % size == front:
            print("Queue is full")
        else:
            x = int(input("Enter element: "))

            if front == -1:
                front = 0

            rear = (rear + 1) % size
            if len(queue) < size:
                queue.append(x)
            else:
                queue[rear] = x

            print("Element inserted")

    elif choice == 2:
        if front == -1:
            print("Queue is empty")
        else:
            print("Deleted element:", queue[front])

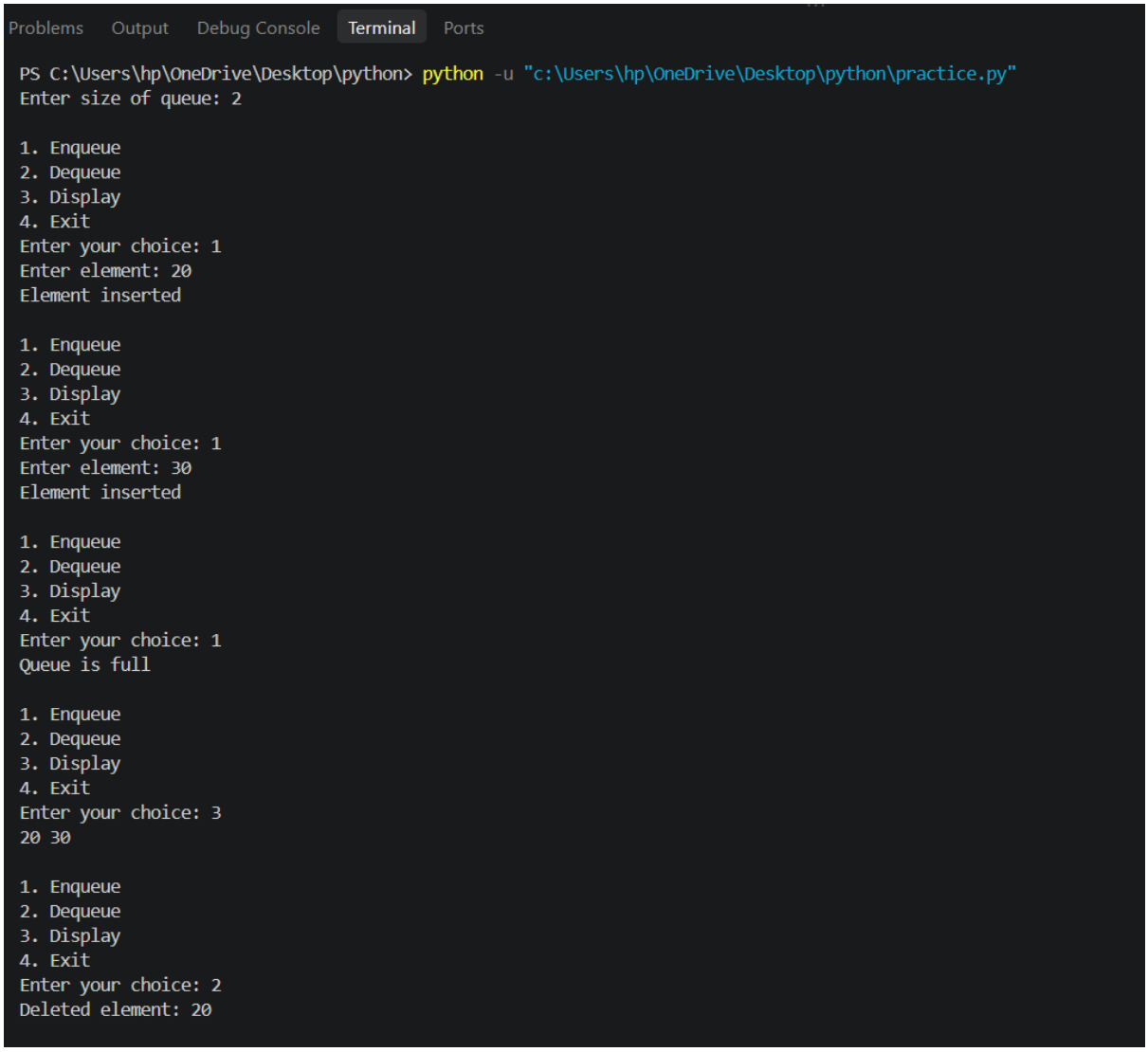
            if front == rear:
                front = rear = -1
            else:
                front = (front + 1) % size

    elif choice == 3:
        if front == -1:
            print("Queue is empty")
        else:
            i = front
            while True:
                print(queue[i], end=" ")
                if i == rear:
                    break
                i = (i + 1) % size
            print()

    elif choice == 4:
        break

    else:
        print("Invalid choice")
```

# OUTPUT


Problems
Output
Debug Console
Terminal
Ports
PS C:\Users\hp\OneDrive\Desktop\python> python -u "c:\Users\hp\OneDrive\Desktop\python\practice.py"
Enter size of queue: 2
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 20
Element inserted
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter element: 30
Element inserted
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Queue is full
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
20 30
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Deleted element: 20